the day will come when we humans set foot on Mars. It is only a matter of when.

According to solemn treaty, signed in Washington and Moscow on January 27, 1967, no nation may lay claim to part or all of another planet. Nevertheless—for historical reasons that Columbus would have understood well—some people are concerned about who first sets foot on Mars. If this really worries us, we can arrange for the ankles of the crew members to be tied together as they alight in the gentle Martian gravity.

The crews would acquire new and previously sequestered samples, in part to search for life, in part to understand the past and future of Mars and Earth. They would experiment, for later expeditions, on extracting water, oxygen, and hydrogen from the rocks and the air and from the underground permafrost—to drink, to breathe, to power their machines and, as rocket fuel and oxidizer, to propel the return voyage. They would test Martian materials for eventual fabrication of bases and settlements on Mars.

And they would go exploring. When I imagine the early human exploration of Mars, it's always a roving vehicle, a little like a jeep, wandering down one of the valley networks, the crew with geological hammers, cameras, and analytic instruments at the ready. They're looking for rocks from ages past, signs of ancient cataclysms, clues to climate change, strange chemistries, fossils, or —most exciting and most unlikely—something alive. Their discoveries are televised back to Earth at the speed of light. Snuggled up in bed with the kids, you explore the ancient riverbeds of Mars.

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*Although in a few places, such as the slopes of the elevation called Alba Patera, there are multibranched valley networks that by comparison are very young. Somehow, even in the most recent billion years, liquid water seems to have flowed here and there, from time to time, through the deserts of Mars.

*Short for Shergotty-Nakhla-Chassigny. You can see why the acronym is used.